

Christof Teuscher

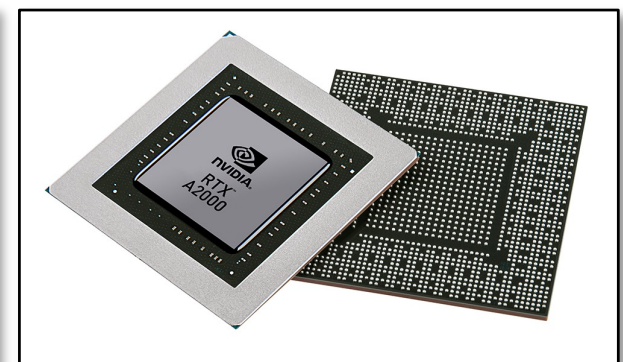
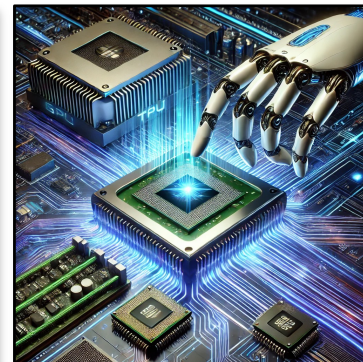
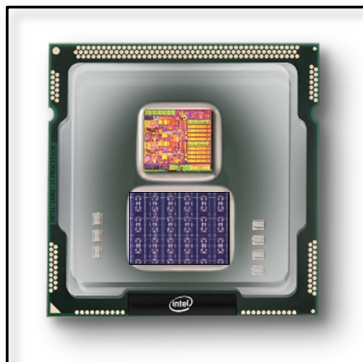
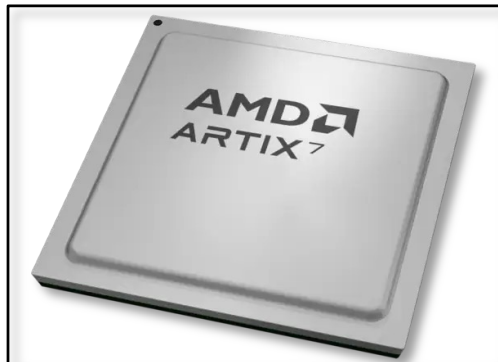
ECE 410/510: Hardware for AI and ML, Spring 2026

Week 8: Recap, outlook, and reminders

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What did you learn last week?



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Analog is not dead

“Analog is dead, long live analog!” — Boris Murmann, Unconventional AI (Apr 2026)

The premise

- Analog wins only at low precision (1–4 bits); each extra bit quadruples energy via $\sqrt{(kT/C)}$ noise.
- Digital 4-bit in-memory MAC arrays already hit >100 TOPS/W. The analog lead is smaller than hoped.

Why TOPS/W misleads

- System metric is energy / inference, not fJ/op; noisy analog forces bigger models & more memory traffic.
- Large LLMs are DRAM-bound, i.e., ~half the system energy is external memory, dwarfing compute gains.

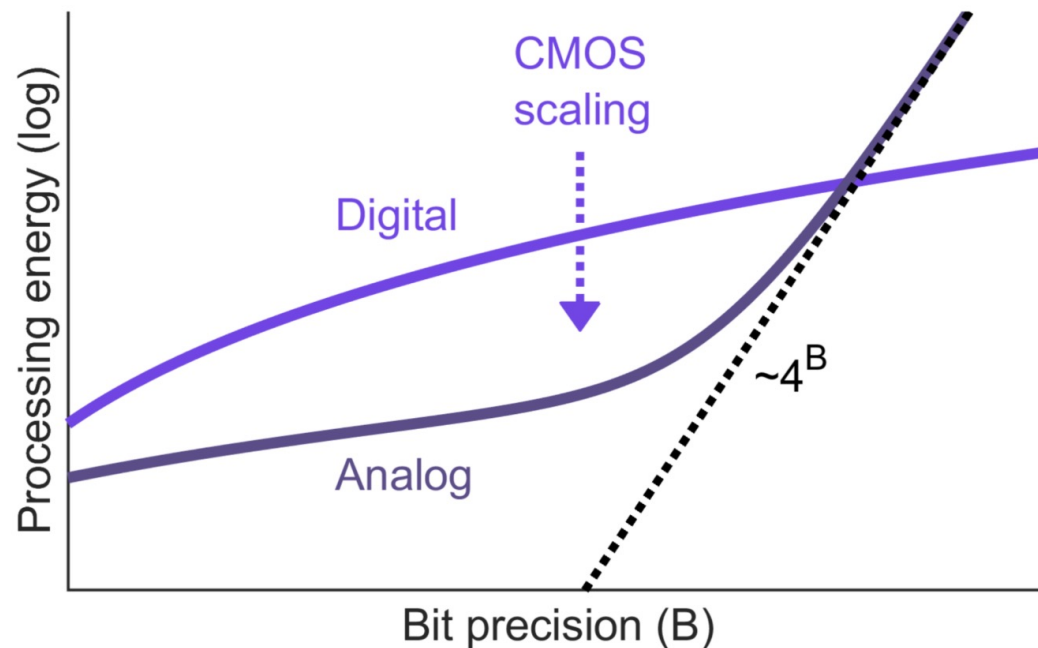
The path forward

- 3D stacking, energy-aware model/**HW co-design**, and alternatives to linear algebra (e.g. dynamical-system state)

Analog is dead inside today’s CMOS linear-algebra stack. But mixed-signal, clean-slate fabrics are where it lives on.

<https://unconv.ai/blog/analog-is-dead-longlive-analog>

Analog is not dead



- 4^B curve is the **theoretical energy cost of achieving B bits of precision** in an analog circuit without any architectural tricks.
- It comes from noise fundamentals: to resolve a signal to B bits, you need a signal-to-noise ratio of at least 2^B , which means you need to suppress noise (thermal, shot, etc.) by that factor.
- **Analog looks energy-efficient at low precision** but runs into this hard physical wall at high precision, where the energy cost grows exponentially.
- **Digital grows much more slowly with precision** (roughly linearly in energy per bit added, since you just add more logic), which is why CMOS scaling has historically favored digital for high-precision computation.

More relevant
than TOPS/W:

$$\frac{\text{Energy}}{\text{Inference}} \approx \left[\frac{\text{Energy}}{\text{Operation}} + \frac{\text{Memory Accesses}}{\text{Operation}} \frac{\text{Energy}}{\text{Memory Access}} \right] \frac{\text{Operations}}{\text{Inference}}$$

Power delivery for DRAM-based PIM

“Power delivery for DRAM-based PIM” — Sundara Raman et al., UT Austin (Apr 2026)

The problem

- DRAM-PIM activates many rows/subarrays in parallel, leading to current spikes the **Power Delivery Network (PDN)** was never designed for
- Bursty, spatially-correlated draw causes voltage drop, **IR (current×resistance) drop**, and thermal hotspots across bank/rank/channel

A unified taxonomy

- Classify PIM current behavior on two axes: **temporal** (burst vs. sustained) and **spatial** (localized vs. distributed)
- Maps subarray-level, bank-level, and 3D-stacked PIM onto the same PDN-stress framework

Mitigations

- Tighten activation timing — **tRRD** (row-to-row activate delay), **tFAW** (four-activate window) — plus PDN-aware scheduling, data placement, and bank/vault power management

DRAM-PIM's throughput gains hinge on PDN-aware **co-design**, not just architecture.

<https://arxiv.org/pdf/2604.04773>

High-accuracy memristor-based analogue CIM

“High-accuracy memristor-based analogue CIM” — Jiang et al., Nature Materials (2026)

The premise

- **Memristor-based analogue CIM offers huge energy and throughput gains, but struggles with computing accuracy.**
- Errors come from device noise & non-idealities across the whole stack, i.e., device, array, and system levels.

Mitigation strategies

- Material/device engineering: more linear, lower-variation switching (e.g. self-selecting molecular memristors).
- Array techniques: peripheral compensation, calibration, redundancy, but each adds hardware/energy overhead.
- Algorithm–architecture co-design: noise-aware training, quantization, mixed-signal partitioning.

Roadmap from prototypes to large-scale deployment hinges on the **accuracy–cost trade-off**.

Sparsity vs pruning

Sparse computation

What it is

- Property of weights/activations: many zeros to skip

Origin

- Designed in (e.g. crossbar-aware) or emerges from ReLU

Hardware angle

- Custom accelerators skip zero MACs and compress memory
- Best when sparsity pattern is co-designed with the HW

Pruning

What it is

- Technique: remove weights / neurons / channels

Origin

- Applied post-training via magnitude or sensitivity criteria

Hardware angle

- Unstructured pruning maps poorly to GPUs/TPUs
- Slide 11 (Liang et al.): crossbar-aware pruning → 44–72% overhead cut

Pruning **creates** sparsity; sparse HW **exploits** it. The two are paired, not competing.

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Crossbar-Aware Neural Network Pruning

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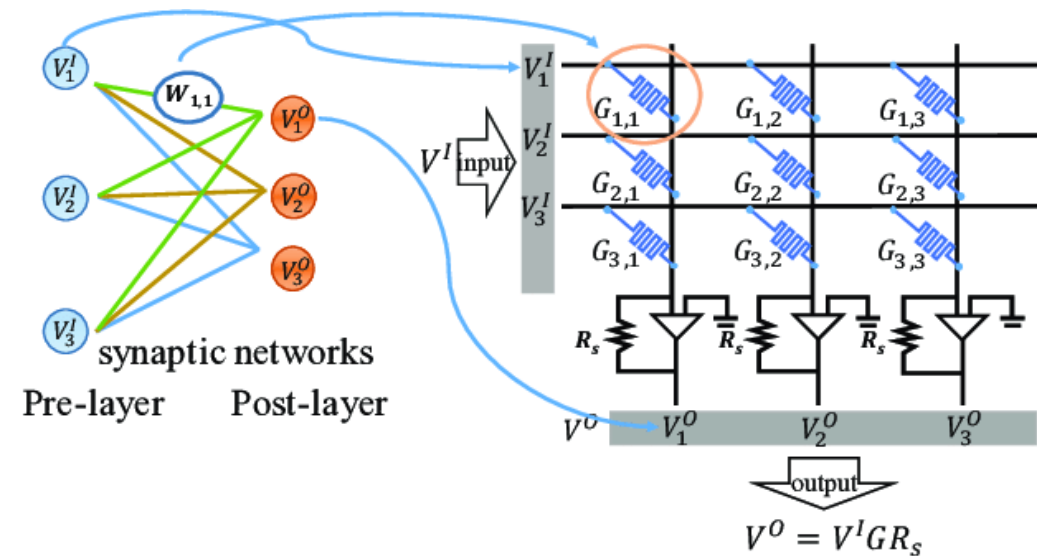
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ABSTRACT Crossbar architecture has been widely adopted in neural network accelerators due to the efficient implementations on vector-matrix multiplication operations. However, in the case of convolutional neural networks (CNNs), the efficiency is compromised dramatically because of the large amounts of data reuse. Although some mapping methods have been designed to achieve a balance between the execution throughput and resource overhead, the resource consumption cost is still huge while maintaining the throughput. Network pruning is a promising and widely studied method to shrink the model size, whereas prior work for CNNs compression rarely considered the crossbar architecture and the corresponding mapping method and cannot be directly utilized by crossbar-based neural network accelerators. This paper proposes a crossbar-aware pruning framework based on a formulated L_0 -norm constrained optimization problem. Specifically, we design an L_0 -norm constrained gradient descent with relaxant probabilistic projection to solve this problem. Two types of sparsity are successfully achieved: 1) intuitive crossbar-grain sparsity and 2) column-grain sparsity with output recombination, based on which we further propose an input feature maps reorder method to improve the model accuracy. We evaluate our crossbar-aware pruning framework on the median-scale CIFAR10 data set and the large-scale ImageNet data set with VGG and ResNet models. Our method is able to reduce the crossbar overhead by 44%–72% with insignificant accuracy degradation. This paper significantly reduce the resource overhead and the related energy cost and provides a new co-design solution for mapping CNNs onto various crossbar devices with much better efficiency.

INDEX TERMS Crossbar architecture, convolutional neural networks, neural network pruning, constrained optimization problem.



Quiz 2

Quiz 2 practice

- Period: May 18 - 24 (week 8)
- Format: oral (on Shine!), closed book, no AI
- Topics: week 5 - 8, lectures + codefests
- Feedback provided

Quiz 2

- Period: May 25 - 31 (week 9)
- Format: oral (on Shine!), closed book, no AI
- Topics: week 5 - 8, lectures + codefests
- Quiz 2 can be retaken (consumes 2 tokens)

Final exam

Final exam practice

- Period: Jun 1 - 7
- Format: oral (on Shine!), closed book, no AI
- Topics: Your project
- Feedback provided

Final exam

- Period: Jun 8 - 14
- Format: oral (on Shine!), closed book, no AI
- Topics: Your project
- The final exam cannot be retaken



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Next week

Monday:

- No class, Memorial Day

Wednesday:

- Codefest 9